

Lecture 1

Lecture 1: Introduction to SDLC

Segment 1: What is Systems Analysis and Design?

Systems Analysis and Design is the process of studying an organization's problems and needs, then creating an information system (IS) to solve those problems and improve the way the organization works.

The main purpose is to help organizations work more efficiently, reduce costs, improve services, and increase profits.

Example

Suppose a university still manages student records using paper files. Finding information takes a lot of time and mistakes often occur. A systems analyst may suggest developing a Student Management System that stores records digitally, making the process faster and more accurate.

Systems Development Life Cycle (SDLC)

The **Systems Development Life Cycle (SDLC)** is a step-by-step process used to develop an information system.

It involves:

1. Determining business needs
2. Designing the system
3. Building the system
4. Testing the system
5. Delivering the system to users
6. Maintaining and improving the system

Example

A hospital wants an online appointment system.

Using SDLC:

- First, analysts identify the hospital's needs.

- Then they design the system.
- Programmers build it.
- The system is tested.
- Doctors, staff, and patients start using it.
- Updates are added when needed.

Role of a Systems Analyst

A **Systems Analyst** plays a key role in the SDLC.

The systems analyst:

- Studies the current business process.
- Finds problems and opportunities for improvement.
- Communicates with users and managers.
- Designs solutions using information systems.
- Helps ensure the new system meets business goals.

Example

A bank experiences long customer waiting times.

The systems analyst investigates the problem and recommends an online banking system that allows customers to transfer money and check balances without visiting the bank.

Why Systems Analysts Challenge Current Processes

Systems analysts do not simply accept the existing way of doing things.

They ask questions such as:

- Why is this process done this way?
- Can it be faster?
- Can it be cheaper?
- Can technology improve it?

Example

A company prints every employee leave request on paper.

A systems analyst may suggest an online leave management system where employees submit requests electronically, saving time and paper.

Failure of Large System Development Projects

Large system development projects often fail.

Research by the **Standish Group** found that for projects with labor costs exceeding **\$100 million**, only about **2%** are completely successful.

Major Reasons for Failure

1. Delays in Decision Making

When managers take too long to make decisions, project progress slows down.

Example:

A company waits several months to approve software requirements, causing development delays.

2. High Workforce Turnover

When experienced employees leave during a project, important knowledge is lost.

Example:

A lead developer leaves halfway through a project, and the new developer needs time to understand the system.

Information System (IS)

An **Information System (IS)** is a combination of people, hardware, software, data, networks, and procedures that work together to collect, process, store, and distribute information.

Most modern software applications are information systems.

Examples include:

- Banking systems
- Hospital management systems
- University management systems
- E-commerce websites
- Airline reservation systems

These systems are commonly developed using programming languages such as:

- Java
- C++
- C#
- .NET technologies

Components of an Information System

An Information System consists of five main components.

1. Hardware

Hardware refers to the physical devices used in a computer system.

Examples:

- Monitor
- Keyboard
- Mouse
- Printer
- Processor (CPU)

Example

In a supermarket billing system, the barcode scanner, computer, and printer are hardware devices.

2. Software

Software consists of programs that tell the hardware what to do.

Examples:

- Windows
- Microsoft Word
- Banking software
- Mobile apps

Example

The software in an ATM machine helps customers withdraw money and check balances.

3. Database

A database is a collection of related data organized for easy access and management.

Examples:

- Student records
- Customer information
- Employee details

Example

A university database stores:

- Student ID
- Name
- Department
- Grades

4. Network

A network connects computers and devices so they can share information and resources.

Examples:

- Internet
- Wi-Fi network
- Office network

Example

All computers in a bank branch are connected through a network so employees can access customer information instantly.

5. Procedures

Procedures are the rules and instructions that explain how people use the information system.

Examples:

- Steps for logging into a system
- Data backup instructions
- Security guidelines

Example

An employee may follow a procedure that requires changing their password every 90 days for security purposes.

Example: Online Shopping System

Component	Example
Hardware	Computer, smartphone, server
Software	Amazon/Daraz application
Database	Customer and product data
Network	Internet connection
Procedures	Steps for placing and processing orders

All five components work together to make the online shopping system function properly.

Systems Analyst and Systems Analysis

Why Do Many Systems Fail?

Many information systems fail because analysts focus on building advanced or impressive systems without first understanding the organization's real needs.

A system is successful only when it solves business problems and creates value for the organization.

Example

A hospital needs a simple patient appointment system. Instead, developers build a complicated system with many unnecessary features. The system becomes expensive, difficult to use, and does not solve the hospital's main problem. As a result, the project may fail.

Primary Goal of a Systems Analyst

The primary goal of a systems analyst is to **create value for the organization**.

Creating value can mean:

- Increasing profits
- Reducing costs
- Improving efficiency
- Saving time
- Improving customer service
- Supporting better decision-making

Example

An online shopping company automates order processing. Employees spend less time entering data manually, which reduces costs and increases efficiency.

Systems Analyst: A Key Person

A systems analyst plays a very important role in system development.

The analyst:

1. Analyzes the Business

Studies how the organization currently works.

2. Identifies Opportunities for Improvement

Finds problems and areas where improvements can be made.

3. Designs Information Systems

Creates information system solutions to implement those improvements.

Example

A university uses paper-based course registration.

The systems analyst:

- Examines the current process.
- Identifies delays and errors.
- Designs an online registration system.

The Systems Analyst

The systems analyst works closely with all project team members to ensure that the right system is developed effectively.

They work with:

- Users
- Managers
- Developers
- Database designers
- Testers

Their responsibility is to ensure the system meets business requirements.

Applying Technology to Business Problems

Systems analysts must understand how technology can solve business problems.

Example

Problem: Customers spend hours waiting to pay utility bills.

Solution: Develop an online payment system.

Technology is used to improve the business process.

Systems Analyst as a Change Agent

A systems analyst is often called a **change agent** because they help organizations adopt new and better ways of working.

They:

- Identify needed improvements.
- Design systems that support those improvements.
- Train users.
- Motivate employees to accept the new system.

Example

A company changes from paper records to a digital database. Employees may resist the change.

The systems analyst:

- Explains the benefits.
- Provides training.
- Helps employees adapt.

Systems Analyst Skills

A systems analyst needs several important skills.

1. Technical Skills

Technical skills help analysts understand technology and system architecture.

They must understand:

- Hardware
- Software
- Databases
- Networks
- System integration

Example

When developing an e-commerce website, the analyst must understand servers, databases, payment gateways, and security mechanisms.

2. Business Skills

Business skills help analysts understand how the organization operates.

They need to know:

- Business processes
- Organizational goals
- Customer needs
- How IT can support business activities

Example

Before designing a sales system, the analyst should understand how products are sold and how inventory is managed.

3. Analytical Skills

Analytical skills help analysts solve problems and make decisions.

They must:

- Gather information

- Analyze situations
- Identify causes of problems
- Evaluate solutions

Example

If customers complain about slow service, the analyst investigates the causes and recommends improvements.

4. Interpersonal Skills

Interpersonal skills help analysts communicate and work effectively with people.

These include:

- Communication skills
- Listening skills
- Teamwork
- Negotiation skills

Example

An analyst interviews employees to understand their requirements. Good communication helps collect accurate information.

5. Management Skills

Management skills help analysts plan and coordinate project activities.

These include:

- Project planning
- Time management
- Resource management
- Leadership

Example

An analyst may create a project schedule and ensure tasks are completed on time.

6. Ethical Skills

Ethical skills help analysts make responsible and fair decisions.

They must:

- Protect confidential information
- Respect user privacy
- Follow laws and regulations
- Use information responsibly

Example

A hospital analyst must keep patient records confidential and secure.

What is Systems Analysis?

Systems Analysis is the process of studying a problem situation and gathering information before designing and implementing an information system.

It uses:

- Notations
- Methodologies
- Tools
- Techniques

to understand business needs and system requirements.

Objectives of Systems Analysis

Systems analysis ensures that the proposed system:

Meets User Needs: The system should solve the users' problems.

Can Be Delivered on Time: Development should be completed within the planned schedule.

Can Be Updated Inexpensively: Future maintenance and modifications should be easy and cost-effective.

Example

Before building a library management system, analysts determine:

- What information must be stored.
- Who will use the system.
- What reports are required.
- What security measures are needed.

Problems in Getting Systems Analysis Right

Systems analysis is often difficult because requirements are not always clear.

1. Ill-Defined Situations

Users may not clearly know or explain what they need.

Example

A manager says: "I want a better system." This requirement is too vague.

2. Ambiguities

Requirements may have more than one meaning.

Example

"The system should be fast."

Questions:

- How fast?
- For which tasks?
- Under what conditions?

3. Inconsistencies

Different users may provide conflicting requirements.

Example

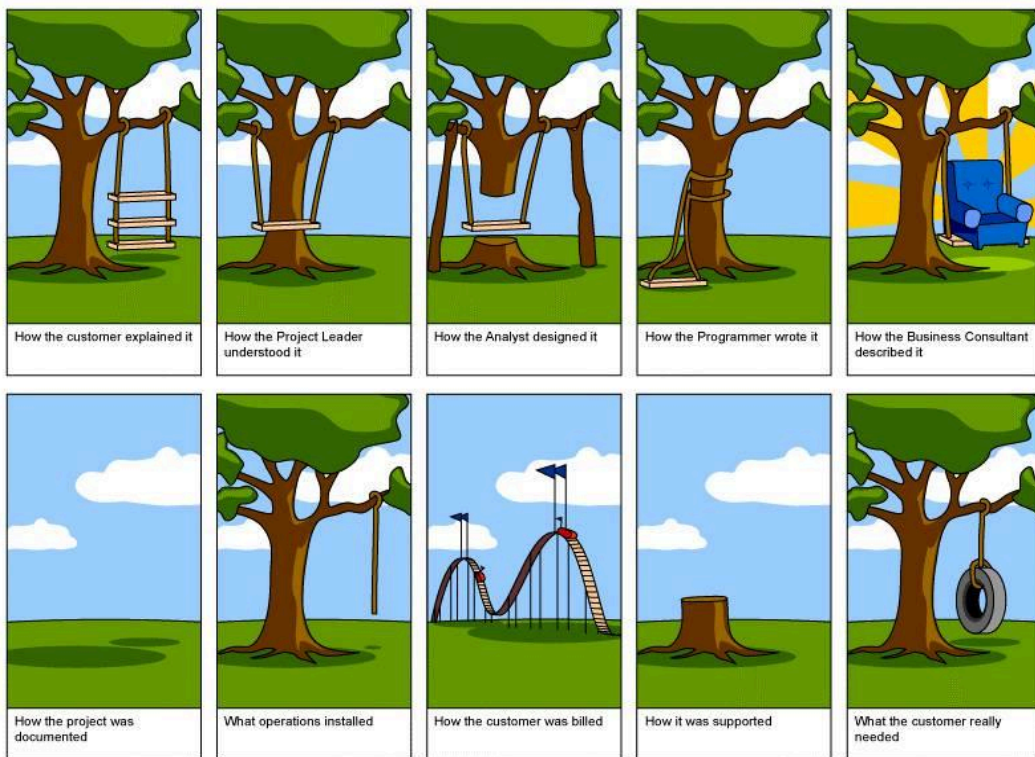
One manager wants strict security. Another manager wants very easy access. These requirements may conflict.

4. Mixing Requirements with Design

Users sometimes suggest technical solutions instead of describing their needs.

Example

The user says: "Use a MySQL database." This is a design choice, not a requirement. The real requirement is: "The system should store customer data securely."



Need for Systems Analysis

Systems analysis is essential because mistakes found later are much more expensive to fix.

A common software engineering principle states:

Fixing a problem after software delivery can cost about 100 times more than fixing it during systems analysis or early design.

Example

During Systems Analysis

An analyst discovers that users need a password recovery feature.

Cost:

- A few hours of work.

After Deployment

Thousands of users are already using the system.

Now the feature requires:

- Additional development
- Testing
- Documentation updates
- User training

The cost becomes much higher.

Segment 2: Software Life Cycle (SDLC)

The **Software Development Life Cycle (SDLC)** is a structured process used to develop an information system from the initial idea to the final working system.

It provides a step-by-step approach to ensure the system is developed properly, on time, and within budget.

Major Attributes of the Life Cycle

A software project has several important characteristics.

1. Moves Through Phases Systematically

The project progresses through a series of phases in a specific order.

Example: Planning → Analysis → Design → Implementation

2. Each Phase Produces Outputs

Every phase creates documents or results called **deliverables**.

Example: The Analysis phase produces a **System Proposal**.

3. Deliverables Are Used in Later Phases

The output of one phase becomes input for the next phase.

Example: Requirements gathered during Analysis are used during Design.

4. Results in an Actual Information System

The final outcome is a working information system.

Example: A completed Hospital Management System.

5. Uses Gradual Refinement

The system becomes more detailed as the project progresses.

Example:

Planning: We need an online library system.

Analysis: Users need book search, issue, and return functions.

Design: Database tables and screen layouts are created.

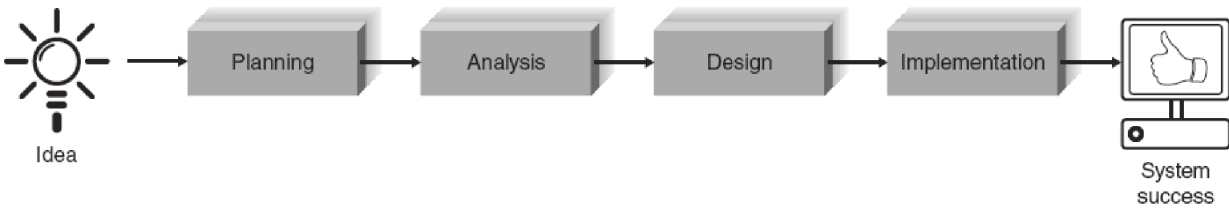
Implementation: Coding and testing are completed.

SDLC Phases

There are four major phases:

1. Planning

2. Analysis
3. Design
4. Implementation



1. Planning Phase

Main Question:

Why should we build the system?

The Planning phase determines whether the project should be undertaken.

Activities in Planning:

Project Initiation

- **System Request.**

A system request contains:

- Business problem
- Business need
- Expected benefits

Example

A university submits a request: "We need an online admission system because manual admissions take too much time."

- **Feasibility Analysis**

Checks whether the project is practical.

Types of feasibility:

- Technical feasibility

- Economic feasibility
- Organizational feasibility

Example Can the university afford the software and hardware costs?

Project Management

Managed by the **Project Manager**.

Responsibilities include:

- **Work Plan**

Defines:

- Tasks
- Schedule
- Deadlines
- **Staff the Project**

Assigns team members.

- **Guide and Control**

Monitors project progress.

Output of Planning

Product: Project Plan

Contains:

- Schedule
- Budget
- Team members
- Resources

Planning Process → Project Plan

The confusion comes from mixing the **main activities** and their **sub-activities**.

2. Analysis Phase

Main Question

- Who will use the system?
- What will the system do?
- When will it be used?
- Where will it be used?

Goal

Understand business requirements and propose a new system.

Activity 1: Determine Analysis Strategy

What happens?

Decide how information will be collected.

Includes:

- Study existing system (As-Is System)
- Identify problems in the current system

Example

Current admission system:

- Paper forms
- Manual verification
- Long delays

Activity 2: Collect and Analyze Requirements

What happens?

Gather user needs and determine what the new system should do.

Includes:

- Requirement gathering
- Develop new system concept (To-Be System)
- Describe new system using analysis models

Requirement Example

Students want:

- Online application
- Online fee payment
- Application status tracking

New System Concept (To-Be System)

- Fully online admission portal

Analysis Models

Business models such as:

- Use Case Diagrams
- DFDs (Data Flow Diagrams)
- Process Models

Activity 3: Prepare and Present System Proposal**What happens?**

Combine all analysis results into one document.

Formula

Analysis + System Concept + Business Models = System Proposal

System Proposal Contains

- Current system problems
- User requirements
- Proposed solution
- Business models

- Expected benefits

Analysis Results

The analyst summarizes everything found during the Analysis Phase.

Output: **System Proposal**

Go / No-Go Decision

The System Proposal is presented to:

- Project Sponsor
- Steering Committee

Decision

GO

→ Move to Design Phase

NO-GO

→ Stop the project or revise it

Complete Flow

Analysis Phase

1. Determine Analysis Strategy

└ Study Existing System & Problems

2. Collect and Analyze Requirements

└ Develop New System Concept

└ Describe New System with Analysis Models

3. Prepare and Present System Proposal

└ Summarize Analysis Results

└─ Go / No-Go Decision

3. Design Phase

Main Question

How will the system work?

Goal

Convert the requirements gathered in the Analysis Phase into a detailed technical design that developers can build.

Activity 1: Determine Design Strategy

What happens?

Decide how the system will be obtained.

Options:

Build: Develop the system using the organization's own developers.

Buy: Purchase an existing software package.

Outsource: Hire an external company to develop the system.

Example

A university may:

- Build its own admission system.
- Buy ready-made admission software.
- Outsource development to a software company.

Activity 2: Design System Components

What happens?

Design all major parts of the system.

Includes

Architecture Design

Defines the overall structure of the system.

Example:

- Web-based system
- Client-server system
- Cloud-based system

Interface Design

Designs how users will interact with the system.

Example:

- Login page
- Student dashboard
- Application form

Database Design

Designs how data will be stored.

Example:

- Student table
- Application table
- Payment table

Program Design

Designs the logic and processing of programs before coding begins.

Example:

- Fee calculation module
- Application verification module

Assemble Design Elements into System Specification

Combine all design components into one document.

System Specification Contains

- Architecture design
- Interface design
- Database design
- Program design
- Technical requirements

Output: System Specification

Activity 3: Present to Steering Committee

The completed System Specification is presented to:

- Steering Committee
- Project Sponsor

Go / No-Go Decision

Before entering the final phase (Implementation), a decision is made.

GO

→ Proceed to the Implementation Phase.

NO-GO

→ Revise the design or stop the project.

Complete Flow

Design Phase

1. Determine Design Strategy

- └ Build / Buy / Outsource

2. Design System Components

- └ Architecture

- └ Interface

- └ Database

- └ Programs

3. Assemble Design Elements

- └ System Specification

4. Present to Steering Committee

- └ Go / No-Go Decision

Output: System Specification

Formula

Architecture + Interface + Database + Programs = System Specification

4.Implementation Phase

Main Question

How will the system be built, installed, and supported?

Goal

Build the system, deliver it to users, and provide ongoing support.

Activity 1: System Construction

What happens?

The actual system is developed.

Includes

Programming: Developers write the program code.

Testing: The system is tested to ensure it works correctly.

Example

For an Online Admission System:

- Create application forms
- Develop payment module
- Develop student dashboard
- Test all features

Activity 2: System Installation

What happens?

The completed system is installed and made available to users.

Includes

Training

Users are trained to use the new system.

Example: University staff learn how to process online applications.

Conversion to New System

The organization switches from the old system to the new system.

Example:

Old System:

- Paper-based admission

New System:

- Online admission portal

Activity 3: On-going System Support

What happens?

Support and maintenance continue after the system is deployed.

Includes

- Fixing errors (bugs)
- Updating features
- Improving performance
- User support
- Security updates

Example

If students cannot upload documents, the support team fixes the issue.

Complete Flow

Implementation Phase

1. System Construction

└ Programming

└ Testing

2. System Installation

└ Training

└ Conversion to New System

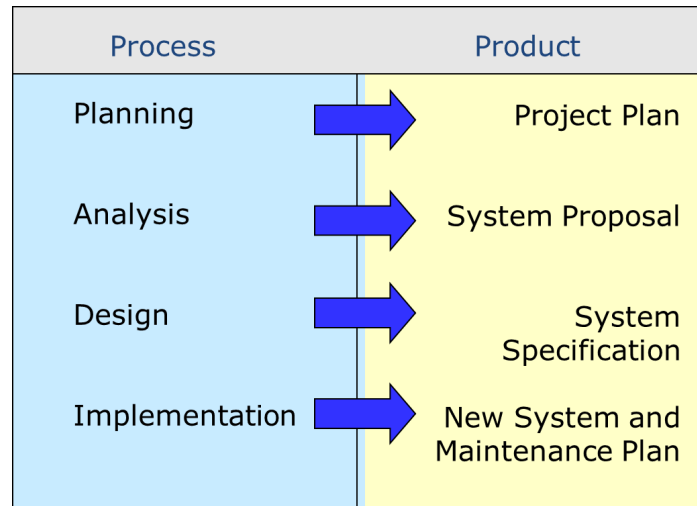
3. On-going System Support

└ Maintenance

└ Updates

└ Bug Fixes

Output: Working Information System



Methodology

A **Methodology** is a formal set of procedures used to carry out the SDLC. It provides guidelines on how to perform each phase.

Categories of Methodologies

1. Process-Centered

Focuses on processes and workflows.

Example

Data Flow Diagram (DFD)-based approaches.

2. Data-Centered

Focuses mainly on data and databases.

Example: Entity Relationship (ER) modeling.

3. Object-Oriented

Focuses on objects that contain both data and behavior.

Example: UML and Java-based development.

Need for Methodology

Without a methodology, large projects become difficult to manage.

Example

Writing code immediately without proper planning may work for a small calculator program.

However, it rarely works for:

- Banking systems
- Hospital systems
- Airline reservation systems

Large systems require:

- Proper requirements
- Proper design
- Proper planning

before implementation starts.

System Development Methodologies

The major SDLC methodologies are:

1. Structured Design: Traditional process-oriented design approach.

Waterfall Development: Phases are completed one after another.

Planning → Analysis → Design → Implementation

Parallel Development: Several parts of the system are developed simultaneously.

2. Rapid Application Development (RAD): Focuses on developing systems quickly.

Phased Development: System is delivered in multiple versions.

Prototyping: Build a working model first to gather feedback.

3. Agile Development: Develops software in small iterations with continuous customer feedback.

Extreme Programming (XP)

An Agile methodology emphasizing:

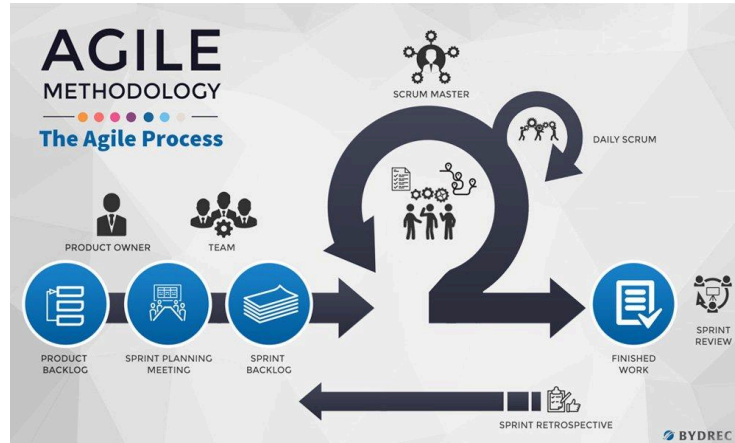
- Continuous testing
- Pair programming
- Frequent releases

4. DevOps

Combines development and operations teams for faster deployment and maintenance.

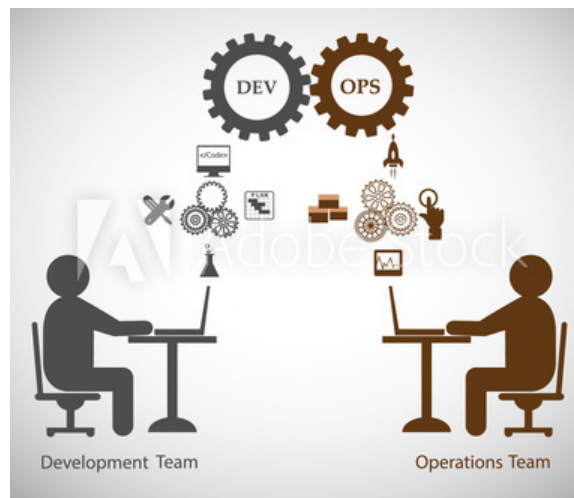
Phase	Main Question	Output
Planning	Why build it?	Project Plan
Analysis	Who, What, When, Where?	System Proposal
Design	How will it work?	System Specification
Implementation	Deliver the system	Working Information System

Segment 3: DevOps



What is DevOps?

DevOps is a **culture and set of practices** that combines Development (Dev) and Operations (Ops) teams to deliver software **faster, more reliably, and with higher quality**.



Why DevOps?

Before DevOps, development and operations teams worked in **separate (isolated) environments**.

- Testing and deployment were done separately after the design-build stage
- These activities took more time than the actual project development
- Teams spent a lot of time in building, testing, designing, and deploying the project
- **Human errors** often happened during manual code deployment
- Development and operations teams had different timelines and were not well coordinated

- This caused delays in project delivery

Because of these problems, there was a need for a better approach like DevOps to improve collaboration and reduce delays.

Current relationship between Development and Operations teams

- The **development team** builds the software and makes sure the code works properly
- After development, they release the code to the **operations team**
- The **operations team** runs the code, manages the system, and checks performance
- If there are any bugs or issues, the operations team reports them back to developers

What DevOps can do?

DevOps helps to solve the gap between the two teams.

- Development and Operations teams work together
- Developers put their code into a **container** (a lightweight software environment)
- When development is finished, this container is given to the operations team
- The operations team runs the container and the application works properly

Example:

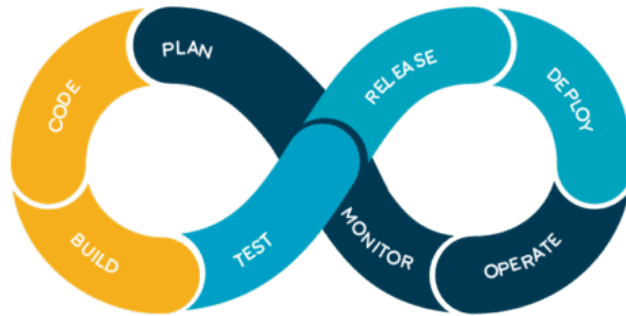
A developer builds a web app and puts it inside a container. The operations team just runs the container without worrying about system setup, and the app works the same everywhere.

Segment 4: DevOps(Cont.)

DevOps is a method where developers (who write code) and operations teams (who manage and run systems) work together. The main goal is to deliver software faster, with better quality, and fewer errors.

Instead of working separately and manually, DevOps uses automation and continuous processes.

DevOps Lifecycle Steps



1. Plan

This is the stage where requirements are collected and decisions are made about what to build. The team studies user feedback and system performance to plan improvements.

Example: Students request an online result system that is faster and easier to use, so the team plans those features.

2. Code

Developers write the actual program using programming languages such as Python, Java, or others.

Example: Developers create code for login, result viewing, and user management in an online system.

3. Build

The written code is converted into a working application. It is compiled and packaged so it can run properly.

Example: The system code is built into an executable web application.

4. Test

The application is tested to find errors or bugs. Testing is often automated using tools like Selenium.

Example: The system is tested to check whether login, search, and result display work correctly. If errors are found, the code is fixed and tested again.

5. Release

The application is prepared for release to users. This step involves planning when and how the new version will be made available.

Example: A new version of an online system is scheduled for release after successful testing.

6. Deploy

The application is installed and made available on the server. Users can now access it through the internet or network.

Example: The result website is uploaded to a cloud server so students can use it.

7. Operate

The system runs in real-world conditions. Users start using the application in daily life.

Example: Students use the online result system to check their results.

8. Monitor

The system is continuously monitored to check performance and detect issues.

Example: If the website becomes slow or crashes, the monitoring system detects it and alerts the team.

Continuous Nature of DevOps

DevOps is not a one-time process. It works continuously.

- Continuous Development: Code is updated regularly
- Continuous Testing: Tests run automatically
- Continuous Integration: Code changes are merged frequently
- Continuous Deployment: Updates are released automatically
- Continuous Monitoring: System performance is always tracked

Without DevOps

- Developer writes code

- Sends it to operations team
- Manual testing is done
- Manual deployment is done
- Problems appear in production

This process takes weeks or even months.

With DevOps

- Developer pushes code
- CI pipeline runs automatically
- Automated testing is performed
- Docker image is created
- Kubernetes deploys the system
- Monitoring tools track performance

Deployment can happen in minutes.

Advantages of DevOps

- Reduces chances of system failure
- Speeds up software delivery
- Improves teamwork between teams
- Increases efficiency
- Provides faster updates and improvements

Challenges of DevOps

- 1. Integration difficulties:** Different tools may not work well together.
- 2. Automated testing complexity:** Setting up proper automated tests can be difficult.
- 3. High cost:** Tools and infrastructure can be expensive.
- 4. Tool selection problem:** There are many DevOps tools, making it hard to choose the right ones.

5. Lack of skilled professionals: Experienced DevOps engineers are not always available.

Segment 5: Project Team Roles and Skills

In a project (especially an information system project), different people have different responsibilities. Each role focuses on a specific part of the system so the project can run smoothly.

Information System Roles

1. Business Analyst

A business analyst understands the business needs and converts them into system requirements.

Main responsibilities:

- Talk to users and stakeholders
- Understand business problems
- Identify what the system should do
- Document requirements clearly

Example:

A university wants an online result system. The business analyst asks students and teachers what features are needed, such as login, result viewing, and grade calculation.

2. System Analyst

A system analyst designs how the system will work technically based on the requirements.

Main responsibilities:

- Analyze system requirements
- Design system structure and workflow
- Work with developers to implement solutions
- Ensure system meets business needs

Example:

After requirements are collected, the system analyst designs how the login system and database will work.

3. Infrastructure Analyst

An infrastructure analyst manages the hardware, networks, servers, and technical environment needed for the system.

Main responsibilities:

- Set up servers and networks
- Ensure system performance and reliability
- Manage cloud or physical infrastructure
- Handle security and backups

Example:

For an online result system, the infrastructure analyst sets up the server so students can access the website without downtime.

4. Change Management Analyst

A change management analyst handles changes in the system during development or after deployment.

Main responsibilities:

- Manage system updates and changes
- Ensure changes do not break the system
- Communicate changes to users and teams
- Control version updates

Example:

If the university adds a new grading system, the change management analyst ensures it is updated without affecting existing data.

5. Project Manager

The project manager leads the entire project and ensures everything is completed on time.

Main responsibilities:

- Plan the project timeline
- Assign tasks to team members
- Monitor progress and budget
- Solve project problems
- Communicate with stakeholders

Example:

The project manager ensures the online result system is completed within the deadline and all teams are working properly.

Project Team Roles

Role	Responsibilities
Business analyst	Analyzing the key business aspects of the system Identifying how the system will provide business value Designing the new business processes and policies
Systems analyst	Identifying how technology can improve business processes Designing the new business processes Designing the information system Ensuring that the system conforms to information systems standards
Infrastructure analyst	Ensuring the system conforms to infrastructure standards Identifying infrastructure changes needed to support the system
Change management analyst	Developing and executing a change management plan Developing and executing a user training plan
Project manager	Managing the team of analysts, programmers, technical writers, and other specialists Developing and monitoring the project plan Assigning resources Serving as the primary point of contact for the project

Lecture 2

Lecture 2: Feasibility Analysis

What is Feasibility Analysis?

Before starting a system development project, an organization must decide:

"Is this project worth doing?"

Feasibility analysis helps determine whether a proposed system:

- Solves a real business problem
- Creates value for the organization
- Is worth the time, money, and effort

The process usually begins with a **System Request**.

Why Are Projects Started?

Projects are not started just because someone wants a new system. They are started to create **business value** using information technology (IT).

Common Business Needs

1. Lower operating costs
2. Increase revenue or sales
3. Improve customer service
4. Improve efficiency
5. Use new or emerging technologies
6. Keep up with competitors
7. Improve access to information

Real-Life Example

A supermarket notices that customers wait too long at checkout counters.

Business Need: Improve customer service and reduce waiting time.

Possible Solution: Introduce a self-checkout system.

Business Value:

- Faster service

- More customer satisfaction
- Reduced staffing costs

System Request

A **System Request** is a document that formally proposes a new system project.

It explains:

1. Why the project is needed
2. What business problem exists
3. How the new system will create business value

Think of it as:

"A proposal asking permission to start a project."

Purpose of a System Request

A System Request:

1. Describes the business problem
2. Explains the expected benefits
3. Provides basic project information
4. Helps management decide whether to approve the project

Who Creates the System Request?

Project Sponsor

The project sponsor prepares the System Request.

Who is the Project Sponsor?

The person who:

1. Recognizes the business need
2. Wants the problem solved

3. Understands the expected business value
4. Supports adoption of new technology
5. Wants the project to succeed

Examples

- CEO
- CIO
- Vice President of Marketing
- Finance Department Manager

Real-Life Example

A marketing manager notices that customer information is scattered across many files.

The manager proposes a Customer Relationship Management (CRM) system.

The marketing manager becomes the **Project Sponsor**.

Approval Committee

After the System Request is prepared, it is sent to the Approval Committee.

What Does the Approval Committee Do?

1. Reviews project proposals
2. Compares projects from different departments
3. Decides which projects deserve company resources

Possible Decisions

1. Approve

The project moves forward.

2. Decline

The project is rejected.

3. Suspend

The project is postponed for future consideration.

Real-Life Example

A company receives three proposals:

1. New HR System
2. New Online Store
3. Inventory Management System

The committee may approve only one because of limited budget.

Identifying Business Value

Business value means the benefit the organization receives from the project.

There are two types:

A. Tangible Value

Benefits that can be measured numerically.

Usually expressed in:

- Money
- Percentage
- Time savings

Examples

1. 2% reduction in operating costs
2. 5% increase in sales
3. \$50,000 annual savings
4. Reduction of processing time from 10 minutes to 5 minutes

Real-Life Example

An automated payroll system reduces payroll processing costs by \$20,000 per year.

This is a **tangible value** because the savings can be measured.

B. Intangible Value

Benefits that are important but difficult to measure precisely.

Examples

1. Better customer service
2. Improved company reputation
3. Better employee satisfaction
4. Stronger competitive position
5. Improved decision making

Real-Life Example

A bank launches a mobile banking app.

Customers become happier and trust the bank more.

Although customer satisfaction is valuable, it is difficult to measure exactly in dollars.

This is an intangible **value**.

Elements of a System Request

A System Request contains five main elements.

1. Project Sponsor

The person or group initiating the project.

Examples

- Finance Department
- Vice President of Marketing
- CIO
- CEO

Example

Sponsor: Marketing Manager

2. Business Need

The business problem or opportunity that motivates the project.

Examples

1. Reach a new market segment
2. Keep up with competitors
3. Improve information access
4. Reduce product defects
5. Improve purchasing processes

Example

An online store wants to attract international customers.

Business Need:

Reach a new market segment.

3. Business Requirements

The capabilities the new system must provide.

Think:

"What should the system be able to do?"

Examples

1. Online access to information
2. Product search feature
3. Customer information collection
4. Performance reporting
5. Online support services

Example

For an online shopping website:

Business Requirements:

1. Product search

2. Online ordering
3. Customer account creation
4. Payment processing

4. Business Value

The benefits expected from the system.

Examples

1. 3% increase in sales
2. 1% increase in market share
3. Reduction of employees by 5 FTEs
4. \$200,000 supply cost savings
5. \$150,000 technology cost savings

Example

Expected Benefits:

- Increase sales by 10%
- Reduce order processing costs by 20%

These are business values.

5. Special Issues or Constraints

Important conditions, restrictions, or requirements that may affect project approval.

Examples

1. Government deadline
2. Holiday season deadline
3. Security clearance requirements
4. Limited budget
5. Legal requirements

Example

A tax reporting system must be completed before the government's filing deadline.

This deadline becomes a special constraint.

Element	Description	Examples
Project Sponsor	The person who initiates the project and who serves as the primary point of contact for the project on the business side	Several members of the finance department Vice president of marketing CIO CEO
Business Need	The business-related reason for initiating the system	Reach a new market segment Offer a capability to keep up with competitors Improve access to information Decrease product defects Streamline supply acquisition processes
Business Requirements	The new or enhanced business capabilities that the system will provide	Provide online access to information Capture customer demographic information Include product search capabilities Produce performance reports Enhance online user support
Business Value	The benefits that the system will create for the organization	3% increase in sales 1% increase in market share Reduction in headcount by 5 FTEs \$200,000 cost savings from decreased supply costs \$150,000 savings from removal of outdated technology

Special Issues or Constraints	Issues that pertain to the approval committee's decision	Government-mandated deadline for May 30 System needed in time for the Christmas holiday season Top-level security clearance needed by project team to work with data
-------------------------------------	---	--

Complete Example of a System Request

Project Sponsor

Sales Manager

Business Need

Customers cannot place orders online, causing lost sales.

Business Requirements

1. Online product catalog
2. Shopping cart
3. Online payment system
4. Order tracking

Business Value

1. Increase sales by 8%
2. Reach customers outside the city
3. Improve customer satisfaction

Special Issues/Constraints

The system must be operational before the holiday shopping season.

System Request—Digital Music Download Project

Project Sponsor: Carly Edwards, Assistant Vice President, Marketing

Business Need: This project has been initiated to increase sales by creating the capability of selling digital music downloads to customers through kiosks in our stores, and over the Internet using our website.

Business Requirements: Using the Web or in-store kiosks, customers will be able to search for and purchase digital music downloads. The specific functionality that the system should have includes the following:

- Search for music in our digital music archive.
- Listen to music samples.
- Purchase individual downloads at a fixed fee per download.
- Establish a customer subscription account permitting unlimited downloads for a monthly fee.
- Purchase music download gift cards.

Business Value: We expect that Tune Source will increase sales by enabling existing customers to purchase specific digital music tracks and by reaching new customers who are interested in our unique archive of rare and hard-to-find music. We expect to gain a new revenue stream from customer subscriptions to our download services. We expect some increase in cross-selling, as customers who have downloaded a track or two of a CD decide to purchase the entire CD in a store or through our website. We also expect a new revenue stream from the sale of music download gift cards.

Conservative estimates of tangible value to the company include the following:

- \$757,500 in sales from individual music downloads
- \$950,000 in sales from customer subscriptions
- \$205,000 in additional in-store or website CD sales
- \$153,000 in sales from music download gift cards

Special Issues or Constraints:

- The marketing department views this as a strategic system. The ability to offer digital music downloads is critical in order to remain competitive in our market niche. Our music archive of rare and hard-to-find music is an asset that is currently underutilized.
- Many of our current loyal customers have been requesting this capability, and we need to provide this service or face the loss of these customers' business.

System Request– Student Record Systems (SRS) Project

Project sponsor: Matt Ferry

Business Need:

Due to increasing registration (expected 5% per year) and a co-related increase in workload with the current manual system, a new system should be created. This will enable us to streamline the registration process, allow students to register for classes themselves, and allow employees to access the system from any computer that has access to an internet connection.

Business Requirements:

- The functionality that the system should have is listed below:
- Maintain records for students enrolled in school
- Maintain courses offered by the school
- Maintain classes offered of the available courses (online and face-to-face)
- Maintain student grades for completed classes
- Allow for Student Self-registration
- Allow staff to work from any location that has an Internet connection and web browser

Business Value:

Conservative estimates of tangible value to the company includes:

- 90% reduction in registration time. (600 hours or \$12,000 instead of 6000 or \$120,000)
- 10% increase in staff productivity (Saving \$50,000) Improved staff morale and student interaction

Special Issues or Constraints:

- Security issues must be addressed, as students will have access to school files.
- Due to rapid increase in enrollment, this project needs to be finished within 1 year

Feasibility Analysis?

After a System Request is submitted, the organization performs a **Feasibility Analysis**.

Its purpose is to answer:

"Should we do this project?"

Feasibility Analysis helps:

1. Decide whether the project should continue.
2. Identify risks before investing time and money.
3. Understand whether the project is practical and beneficial.

Simple Idea

Having a good project idea does not automatically mean the project should be approved.

The organization must check:

1. Can we build it?
2. Should we build it?
3. Will people accept and use it?

These questions lead to three types of feasibility.

Three Types of Feasibility

1. Technical Feasibility

Checks whether the organization has the technical ability to develop and run the system.

2. Economic Feasibility

Checks whether the benefits are greater than the costs.

3. Organizational Feasibility

Checks whether employees, managers, and stakeholders will accept the system.

Feasibility Type	Main Question
Technical Feasibility	Can we build it?
Economic Feasibility	Should we build it?

Organizational Feasibility	Will people support and use it?
----------------------------	---------------------------------

Technical Feasibility

Main Question

Can the organization successfully build the proposed system?

Technical feasibility focuses on technical capability, resources, and complexity.

Factor 1: Familiarity with Application

Does the organization understand the business area where the system will be used?

Examples

- Banking system
- Hospital management system
- Online shopping system

Real-Life Example

A software company has already built several hospital systems.

They understand:

- Patient records
- Appointments
- Billing

Therefore, technical feasibility is higher.

Low Familiarity Example

A company that only develops educational software suddenly wants to build an airline reservation system.

They lack domain knowledge.

Technical risk becomes higher.

Factor 2: Familiarity with Technology

Does the organization already know the technology needed?

Examples

- Java
- Python
- Cloud computing
- AI systems
- Mobile applications

High Familiarity Example

The company already develops web applications and wants another web application.

Existing knowledge can be reused.

Technical feasibility is high.

Low Familiarity Example

The company wants to use Artificial Intelligence for the first time.

Employees have no AI experience.

Technical feasibility decreases.

Factor 3: Project Size

How large and complex is the project?

Consider:

1. Number of people needed
2. Development time
3. Number of features
4. Project complexity

Small Project Example

Employee attendance system.

- 4 developers
- 3 months
- Few features

Risk is relatively low.

Large Project Example

National digital healthcare platform.

- Hundreds of developers
- Multiple years
- Thousands of requirements

Risk is much higher.

Factor 4: Compatibility

Can the new system work with existing company technology?

High Compatibility Example

New payroll software works directly with the existing employee database.

Easy integration.

Technical feasibility is high.

Low Compatibility Example

New software cannot communicate with existing databases.

Major modifications are needed.

Technical feasibility is lower.

Technical Feasibility Summary

Ask these questions:

1. Do we understand the business area?
2. Do we know the required technology?
3. Is the project size manageable?
4. Can it work with existing systems?

If most answers are "yes", technical feasibility is high.

Economic Feasibility

Main Question

Should we build it?

Even if a project can be built, it may not be worth the cost.

Economic feasibility compares:

Benefits vs Costs

Goal

Determine whether the project creates enough value to justify the investment.

Cost-Benefit Analysis

The main tool used in economic feasibility.

Steps

Step 1: Identify Costs

What will the project cost?

Step 2: Identify Benefits

What benefits will the project provide?

Step 3: Assign Values

Convert costs and benefits into monetary values whenever possible.

Step 4: Calculate

- Cash Flow
- Return on Investment (ROI)

Then decide whether the project is worthwhile.

5. Development Costs

Meaning

Costs incurred while creating the system.

These are one-time costs.

Examples

1. Programmer salaries
2. Analyst salaries
3. Hardware purchases
4. Software licenses
5. Training costs
6. Consulting fees

Real-Life Example

Building an online shopping system:

Item	Cost
Developers	\$40,000
Database Software	\$10,000
Testing	\$5,000
Training	\$5,000

Total Development Cost = \$60,000

Operational Costs

Costs incurred after the system is deployed and running.

These are recurring costs.

Examples

1. Maintenance
2. Technical support
3. Cloud hosting
4. Internet services
5. Software subscription fees
6. Employee salaries

Example

Annual operating expenses:

Item	Cost per Year
Cloud Hosting	\$8,000
Support Staff	\$12,000
Maintenance	\$5,000

Total = \$25,000/year

Tangible Benefits

Benefits that can be measured directly in money.

Examples

1. Increased sales
2. Reduced labor costs
3. Reduced inventory costs
4. Lower maintenance expenses
5. Faster processing

Real-Life Example

Online ordering system increases annual sales by \$100,000.

This is a tangible benefit because it can be measured.

Intangible Benefits

Benefits that are important but difficult to measure financially.

Examples

1. Better customer satisfaction
2. Improved company image
3. Better decision making
4. Improved employee morale
5. Stronger competitive position

Real-Life Example

Customers become happier because the website is easier to use.

This is valuable but difficult to express exactly in dollars.

Therefore it is an intangible benefit.

Example: Economic Feasibility Calculation

Suppose a company wants a new online ordering system.

Costs

Development Cost = \$50,000

Annual Operating Cost = \$10,000

Benefits

Annual Increased Sales = \$80,000

Annual Cost Savings = \$20,000

Total Annual Benefit = \$100,000

Result

Annual Benefit = \$100,000

Annual Cost = \$10,000

Initial Cost = \$50,000

Benefits greatly exceed costs.

Therefore:

Economic Feasibility = High

The project is financially attractive.

Economic Feasibility Steps

1. Identify Costs and Benefits

List the tangible costs and benefits for the project. Include both one-time and recurring costs.

2. Assign Values to Costs and Benefits

Work with business users and IT professionals to create numbers for each of the costs and benefits. Even intangibles should be valued if at all possible.

3. Determine Cash Flow

Forecast what the costs and benefits will be over a certain period, usually, three to five years. Apply a growth rate to the values, if necessary.

4. Assess Project's Economic Value

Evaluate the project's expected returns in comparison to its costs. Use one or more of the following evaluation techniques:

- **Return on Investment (ROI)**

Calculate the rate of return earned on the money invested in the project, using the ROI formula.

- **Break-Even Point (BEP)**

Find the year in which the cumulative project benefits exceed cumulative project costs. Apply the breakeven formula, using figures for that year. This calculation measures how long it will take for the system to produce benefits that cover its costs.

- **Net Present Value (NPV)**

Restate all costs and benefits in today's dollar terms (present value), using an appropriate discount rate. Determine whether the total present value of benefits is greater than or less than the total present value of costs.

Example Cost & Benefits

Development Costs	Operational Costs
Development Team Salaries	Software Upgrades
Consultant Fees	Software Licensing Fees
Development Training	Hardware Repairs
Hardware and Software	Hardware Upgrades
Vendor Installation	Operational Team Salaries
Office Space and Equipment	Communications Charges
Data Conversion Costs	User Training
Tangible Benefits	Intangible Benefits
Increased Sales	Increased Market Share
Reductions in Staff	Increased Brand Recognition
Reductions in Inventory	Higher Quality Products
Reductions in IT Costs	Improved Customer Service
Better Supplier Prices	Better Supplier Relations

Assign Values

Benefits^a

Increased sales	500,000
Improved customer service ^b	70,000
Reduced inventory costs	68,000
Total benefits	638,000

Development costs

2 servers @ \$125,000	250,000
Printer	100,000
Software licenses	34,825
Server software	10,945
Development labor	1,236,525
Total development costs	1,632,295

Operational costs

Hardware	54,000
Software	20,000
Operational labor	111,788
Total operational costs	185,788

Total costs	1,818,083
--------------------	------------------

^a An important yet intangible benefit will be the ability to offer services that our competitors currently offer.

^b Customer service numbers have been based on reduced costs for customer complaint phone calls.

Cost-Benefit Analysis

	2012	2013	2014	2015	2016	Total
Benefits						
Increased sales		500,000	530,000	561,800	595,508	2,187,308
Reduction in customer complaint calls ^a		70,000	70,000	70,000	70,000	280,000
Reduced inventory costs		68,000	68,000	68,000	68,000	272,000
Total Benefits^b		638,000	668,000	699,800	733,508	2,739,308
Development Costs						
2 servers @ \$125,000	250,000	0	0	0	0	250,000
Printer	100,000	0	0	0	0	100,000
Software licenses	34,825	0	0	0	0	34,825
Server software	10,945	0	0	0	0	10,945
Development labor	1,236,525	0	0	0	0	1,236,525
Total Development Costs	1,632,295	0	0	0	0	1,632,295
Operational Costs						
Hardware		50,000	50,000	50,000	50,000	200,000
Software		20,000	20,000	20,000	20,000	80,000
Operational labor		115,000	119,600	124,384	129,359	488,343
Total Operational Costs		185,000	189,600	194,384	199,359	768,343
Total Costs	1,632,295	185,000	189,600	194,384	199,359	2,400,638
Total Benefits – Total Costs	(1,632,295)	453,000	478,400	505,416	534,149	338,670
Cumulative Net Cash Flow	(1,632,295)	(1,179,295)	(700,895)	(195,479)	338,670	
Return on Investment (ROI)	14.1%	(338,670/2,400,638)				
Break-even Point	3.37 years	(3 years of negative cumulative cash flow + [534,149 – 338,670]/534,149 = .37)				

^a Customer service values are based on reduced costs of handling customer complaint phone calls.

^b An important yet intangible benefit will be the ability to offer services that our competitors currently offer.

Return on Investment (ROI)

What is ROI?

ROI measures how much profit or benefit you get compared to the amount of money you invested.

In simple words:

"For every dollar (or taka) invested, how much do I earn back?"

Higher ROI is better because it means the project's benefits are much greater than its costs.

ROI Formula

$$\text{ROI} = ((\text{Total Benefits} - \text{Total Costs}) \div \text{Total Costs}) \times 100\%$$

Example 1

Suppose:

- Total Benefits = \$150,000
- Total Costs = \$100,000

Step 1

$$\text{Profit} = \text{Benefits} - \text{Costs}$$

$$= 150,000 - 100,000$$

$$= 50,000$$

Step 2

$$\text{ROI} = (50,000 \div 100,000) \times 100$$

$$= 50\%$$

Answer

$$\text{ROI} = 50\%$$

This means for every \$1 invested, the project earns an additional \$0.50 profit.

Example 2

Suppose:

- Benefits = \$500,000
- Costs = \$400,000

$$\text{ROI} = (500,000 - 400,000) \div 400,000 \times 100$$

$$\text{ROI} = 25\%$$

Interpreting ROI

ROI	Meaning
Positive ROI	Benefits exceed costs
Zero ROI	Benefits equal costs
Negative ROI	Costs exceed benefits
Higher ROI	More attractive project

ROI Calculation

	Year 0	Year 1	Year 2	Year 3	Total
Total Benefits		45,000	50,000	57,000	152,000
Total Costs	100,000	10,000	12,000	16,000	138,000
Net Benefits (Total Benefits – Total Costs)	(100,000)	35,000	38,000	41,000	14,000
Cumulative Net Cash Flow	(100,000)	(65,000)	(27,000)	14,000	

$$\text{ROI} = \frac{\text{Total Benefits} - \text{Total Costs}}{\text{Total Costs}}$$

$$\text{ROI} = \frac{152,000 - 138,000}{138,000} = \frac{14,000}{138,000} = 10.14\%$$

Break-Even Point (BEP)

What is Break-Even Point?

Break-Even Point is the time at which:

Total Benefits = Total Costs

At this point:

- No profit
- No loss

The organization has recovered all invested money.

Main Idea

Before BEP:

- Costs are greater than benefits
- Project is still recovering investment

After BEP:

- Benefits become greater than costs
- Project starts generating profit

Why is BEP Important?

Short BEP

Example: 1 year

- Less risk
- Faster recovery
- More attractive project

Long BEP

Example: 8 years

- More risk
- Slower recovery
- Greater uncertainty

Therefore:

Greater BEP → Greater Risk

Graph Interpretation

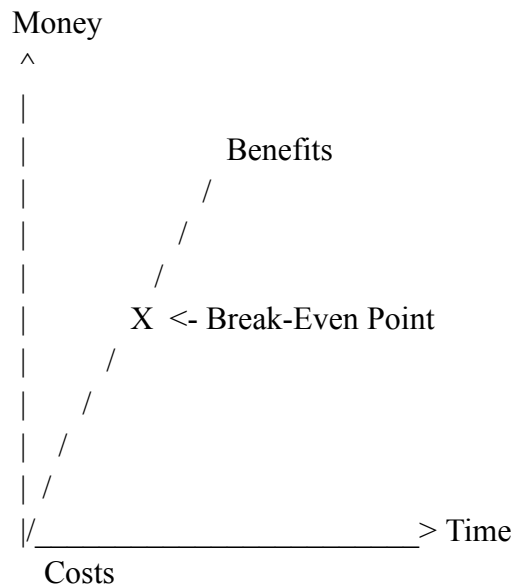
The textbook often explains BEP using a graph.

Draw:

1. Cumulative Costs line
2. Cumulative Benefits line

The point where the two lines intersect is the:

Break-Even Point



At X:

Total Benefits = Total Costs

	Year 0	Year 1	Year 2	Year 3	Total
Total Benefits		45,000	50,000	57,000	152,000
Total Costs	100,000	10,000	12,000	16,000	138,000
Net Benefits (Total Benefits – Total Costs)	(100,000)	35,000	38,000	41,000	14,000
Cumulative Net Cash Flow	(100,000)	(65,000)	(27,000)	14,000	

Number of years of negative cash flow + $\frac{\text{That year's Net Cash Flow} - \text{That year's Cumulative Cash Flow}}{\text{That year's Net Cash Flow}}$

BEP = $2 + \frac{41,000 - 14,000}{41,000} = 2 + \frac{28,000}{41,000} = 2.68 \text{ years}$

How to calculate Number of years of negative cash flow?

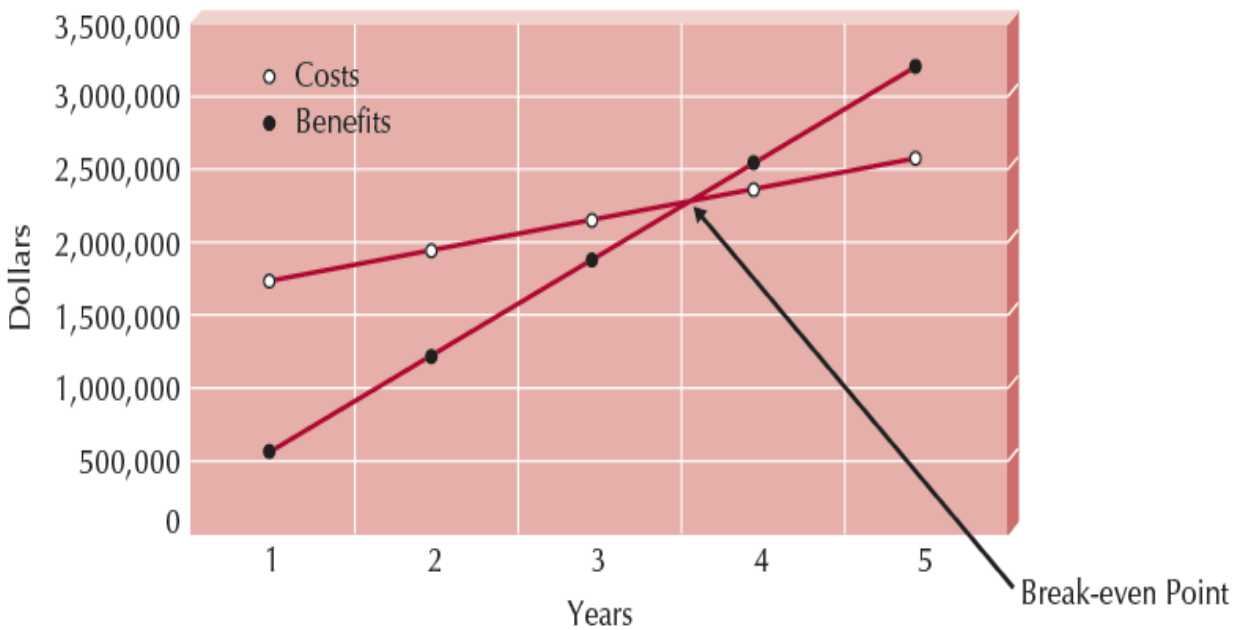
Step 1: Find the number of cumulative net cash flow, N [here $N = 3$]

Step 2: Look at initial year Total benefit.

if benefit present then N unchanged

if No benefit then $N = N - 1$ [No benefit so $N = 3 - 1 = 2$]

Break-Even Graph



	2022	2023	2024	2022	2025	Total
Benefits						
Increased sales		500,000	530,000	561,800	595,508	
Reduction in customer complaint calls ^a		70,000	0	0	70,000	
Reduced inventory costs		68,000	70,000	70,000	68,000	
			68,000	68,000		
Total Benefits^b		638,000	668,000	699,800	733,508	
Present Value Total Benefits		601,887	594,518	587,566	581,007	2,364,978
Development Costs						
2 servers @ \$125,000		0	0	0	0	
Printer	250,000	0	0	0	0	
Software licenses		0	0	0	0	
Server software	100,000	0	0	0	0	
Development labor		0	0	0	0	
	34,825					
	10,945					
	1,236,5					

	25					
Total Development Costs	1,632,295	0	0	0	0	
Operational Costs						
Hardware		50,000			50,000	
Software		20,000	50,000	50,000	20,000	
Operational labor		115,000	20,000	20,000	129,359	
			119,600	124,384		
Total Operational Costs		185,000	189,600	194,384	199,359	
Total Costs	1,632,295	185,000	189,600	194,384	199,359	
Present Value Total Costs	1,632,295	174,528	168,743	163,209	157,911	2,296,686
NPV (PV Total Benefits – PV Total Costs)						68,292

*6% interest rate

Present Value Calculation

PRESENT VALUE EQUALS
Cash flow amount
Divided by
$(1 + \text{interest rate})^n$
Where "n" equals the number of periods

\$100 received in 3 years with a required rate of return of 10% has a PV of \$75.13.

$$PV = \frac{100}{(1 + .10)^3} = \frac{100}{1.331} = 75.13$$

	Year 0	Year 1	Year 2	Year 3	Total
Total Benefits		45,000	50,000	55,000	
PV of Total Benefits		40,909	41,322	42,825	125,056
Total Costs	100,000	10,000	12,000	16,000	
PV of Total Costs	100,000	9,091	9,917	12,021	131,029

Net Present Value (NPV)

The NPV is simply the difference between the total present value of the benefits and the total present value of the costs.

$$\begin{aligned} NPV &= \sum PV \text{ of Total Benefits} - \sum PV \text{ of Total Costs} \\ &= \$125,056 - \$131,029 = (\$5,973) \end{aligned}$$

Formulas

Calculation	Definition	Formula
Present Value (PV)	The amount of an investment today compared to that same amount in the future, taking into account inflation and time.	$\frac{\text{Amount}}{(1 + \text{interest rate})^n}$ n = number of years in future
Net Present Value (NPV)	The present value of benefit less the present value of costs.	PV Benefits – PV Costs
Return on Investment (ROI)	The amount of revenues or cost savings results from a given investment.	$\frac{\text{Total benefits} - \text{Total costs}}{\text{Total costs}}$
Break-Even Point	The point in time at which the costs of the project equal the value it has delivered.	$\frac{\text{Yearly NPV}^* - \text{Cumulative NPV}}{\text{Yearly NPV}^*}$
*Use the Yearly NPV amount from the first year in which the project has a positive cash flow. Add the above amount to the year in which the project has a positive cash flow.		

Organizational Feasibility

Main Question

"If we build it, will they come?"

In simple words:

"If we develop the system, will people actually accept it and use it?"

A technically perfect and economically profitable system can still fail if employees refuse to use it.

Therefore, organizational feasibility checks whether the organization is ready and willing to adopt the new system.

What Does Organizational Feasibility

Examine?

It focuses on two major questions:

1. Will the users accept the system?

Will employees be willing to use the new system?

2. Will the system become part of the organization?

Can the system be successfully integrated into daily business operations?

Real-Life Example

Suppose a company introduces a new attendance system based on facial recognition.

Technical Feasibility

Yes, the technology works.

Economic Feasibility

Yes, it reduces administrative costs.

Organizational Problem

Employees dislike it because they think it invades privacy.

Result:

Many employees resist using it.

The project may fail despite being technically and economically feasible.

This is an **organizational feasibility issue**.

How to Assess Organizational Feasibility?

There are two major activities:

1. Check Strategic Alignment

Strategic Alignment means checking whether the project supports the organization's goals and business strategy.

Simple Question

"Does this project help the company achieve its objectives?"

High Strategic Alignment

The project directly supports company goals.

Example

Company Strategy:

Become the leading online retailer.

Proposed Project:

Develop an advanced e-commerce website.

Result:

Strong strategic alignment.

The project supports the company's strategy.

Low Strategic Alignment

The project does not contribute much to business goals.

Example

Company Strategy:

Expand international sales.

Proposed Project:

Replace office wallpaper software.

Result:

Poor strategic alignment.

The project provides little support for the strategic goal.

2. Perform Stakeholder Analysis

What is a Stakeholder?

A stakeholder is:

Any person, group, or organization that can affect the system or will be affected by the system.

Simple Rule

Ask:

"Who can influence the project?"

or

"Who will be impacted by the project?"

Anyone fitting either category is a stakeholder.

Why is Stakeholder Analysis Important?

Different stakeholders may:

- Support the project
- Oppose the project
- Influence project success
- Influence project failure

Understanding their opinions helps manage resistance and gain support.

Main Stakeholders Considered

1. Project Champion(s)

People who strongly support and promote the project.

They act as the project's biggest supporters.

Responsibilities

1. Encourage the project
2. Obtain resources
3. Convince others to support it
4. Remove obstacles

Example

The Sales Director strongly supports a new CRM system and continually pushes management to approve it.

The Sales Director is the **Project Champion**.

2. Organizational Management

Managers and executives responsible for organizational decisions.

Why Important?

Management:

- Approves budgets
- Allocates resources
- Sets priorities
- Influences employees

Example

A CEO supports a digital transformation initiative.

Employees are more likely to accept the project because management supports it.

3. System Users

People who will use the system in their daily work.

Why Important?

They determine whether the system succeeds in practice.

Even the best system can fail if users reject it.

Example

A hospital introduces a new patient-record system.

System Users:

- Doctors
- Nurses
- Receptionists

If these users find the system difficult and refuse to use it, the project may fail.

Stakeholder Analysis Example

Proposed System

Online Student Registration System

Project Champion

University Registrar

Reason:

Wants registration to be faster and more efficient.

Organizational Management

Vice Chancellor and Administration

Reason:

Need better operational efficiency.

System Users

1. Students
2. Academic Advisors
3. Administrative Staff

Reason:

They will use the system every day.

Signs of High Organizational Feasibility

1. Project supports business strategy.
2. Management supports the project.
3. Users are willing to use the system.
4. Major stakeholders support the project.

5. Minimal resistance to change.

Signs of Low Organizational Feasibility

1. Users resist the system.
2. Management is not supportive.
3. Project conflicts with company goals.
4. Stakeholders oppose the project.
5. Significant organizational changes are required.

Stakeholders

	Role	Techniques for improvement
Champion	A champion: • Initiates the project • Promotes the project • Allocates his or her time to project • Provides resources	• Make a presentation about the objectives of the project and the proposed benefits to those executives who will benefit directly from the system • Create a prototype of the system to demonstrate its potential value
Organizational Management	Organizational managers: • Know about the project • Budget enough money for the project • Encourage users to accept and use the system	• Make a presentation to management about the objectives of the project and the proposed benefits • Market the benefits of the system using memos and organizational newsletters • Encourage the champion to talk about the project with his or her peers
System Users	Users: • Make decisions that influence the project • Perform hands-on activities for the project • Ultimately determine whether the project is successful by using or not using the system	• Assign users official roles on the project team • Assign users specific tasks to perform with clear deadlines • Ask for feedback from users regularly (e.g., at weekly meetings)

Digital Music Download Project Executive Summary

Carly Edwards and Jason Wells created the following feasibility analysis for the Tune Source Digital Music Download Project. The System Request is attached, along with the detailed feasibility study. The highlights of the feasibility analysis are as follows:

Technical Feasibility

The Digital Music Download system is feasible technically, although there is some risk.

Tune Source's risk regarding familiarity with music download applications is moderately high.

- The Marketing Department has little experience with a subscription-based business model.
- The IT department has strong knowledge of the company's existing Web-based CD sales system, but it has not worked with music downloads or customer subscriptions.
- Numerous music download sites exist on the Internet.

Tune Source's risk regarding familiarity with the technology is moderately low.

- The IT department has knowledge of the current Web-based order entry system and the databases and Internet technology it uses.
- The IT department has no direct knowledge of the technology required to store and deliver digital music downloads; however, many of the technical issues will be the responsibility of the ISP.
- Consultants are readily available to provide help in this area.

The project size is considered medium risk.

- The project team will likely consist of 10 or fewer people.
- Business user involvement will be required.
- The project time frame is somewhat critical, since the system is needed to maintain our competitive position in the market.

The compatibility with Tune Source's existing technical infrastructure should be good.

- An Internet infrastructure is already in place at the retail stores and corporate headquarters.
- The ISP should be able to scale its services to accommodate the new Digital Music Download system.

Economic Feasibility

A cost-benefit analysis was performed; see attached spreadsheet for details (provided in Appendix 1A). Conservative estimates show that the Digital Music Download system has a good chance of significantly enhancing the company's bottom line.

ROI over 3 years: 280%

NPV over 3 years: \$4,180,431

Break-even occurs after 0.17 years

Intangible Costs and Benefits

Improved customer satisfaction.

Enhanced competitive position through expansion of our brand into the music download market.

Organizational Feasibility

From an organizational perspective, this project has low risk. The top executives of the company have a strong interest in the project, and the project champion, Carly Edwards, is a respected and knowledgeable marketing executive.

The users of the system, Internet consumers and in-store kiosk users, are expected to appreciate the entry of Tune Source into the music download arena. Management at the stores may have some concern about lost CD sales; however, since customers have so many other options available for music downloads, this system may prevent our losing those customers to other digital music sources and may provide us with the opportunity to cross-sell those customers from our CD inventory.

Additional comments:

- The Marketing Department views this as a strategic system. This system will allow us to leverage our music archive and our well-established market position to establish a presence in the digital music download business. Our customers have been requesting such a capability, and we believe it will be well accepted.
- We should consider hiring a consultant with expertise in similar applications to assist with the project.
- We will need new staff to operate the system and potentially to provide customer service for subscribers and gift-card holders.